SIMATS ENGINEERING

DEPARTMENT OF INFORMATION TECHNOLOGY

ITA05 – COMPUTER VISION

ASSIGNMENT 1 – CO1

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Faculty Name	D Murali	**Nature of Assignment**	Individual
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Assignment Title

Apply, Compare and Justify Image Acquisition, Formation and Processing Techniques for a Textile Fabric Defect Inspection System

Industrial Scenario

Sri Lakshmi Textile Industries, Tiruppur, plans to install a computer-vision inspection system on a moving fabric production line. It must inspect cotton and cotton-polyester fabric, measure fabric width, detect holes, broken threads, oil stains, weaving irregularities and colour variations, locate each defect, and trigger a marking mechanism for rework or rejection.

Given Industrial Data and Constraints

Parameter	Given value
Fabric width	1,000 mm
Camera field of view	1,200 × 400 mm
Conveyor speed	0.6 m/s
Smallest defect	1.5 mm
Oil-stain contrast	5– 8 grey levels
Width accuracy	±2 mm
Maximum decision time	0.5 s
Camera-to-marker distance	0.3 m
Lighting variation	500– 3,000 lux
Camera height	700 mm
Computer	4-core CPU, 16 GB RAM, no GPU
Software	Python, OpenCV and NumPy
Network	Gigabit Ethernet
Storage	2 TB
Budget	₹ 1,50,000
Image retention	7 days
Maximum missed-defect rate	2%

Note: Cost and performance figures in Task 9 are planning estimates, not vendor quotations. Final performance must be confirmed through trials on actual fabric.

TASK 1 – Overview and Features of Computer Vision

1.1 Working principles

Approach	Working principle
Manual inspection	Trained workers observe the moving fabric under suitable light and visually identify defects. They mark or record defective portions.
Image-processing-assisted inspection	A camera captures fabric images. Software enhances the images and highlights suspected defects. An operator checks the results and confirms whether they are real defects.
Fully automated computer vision	A camera and controlled light continuously capture the fabric. Algorithms detect and classify defects, calculate their coordinates and send a signal to the marking mechanism without routine operator review.

1.2 Comparison

Factor	Manual	Assisted	Fully automated
Accuracy	Depends on worker skill and fatigue	More repeatable; operator can reject false alarms	Potentially high after validation; depends on lighting, calibration and algorithm
Speed	Limited by human observation	Faster screening, but operator review adds delay	Suitable for high-speed continuous inspection
Consistency	May vary between workers and shifts	More consistent image analysis	Consistent rules and thresholds
Labour	High	Medium	Low routine labour; maintenance still needed
Continuous production	Fatigue can reduce performance	Useful as a transition system	Best suited to uninterrupted production

1.3 Selection and justification

The fully automated computer-vision approach is recommended. The line moves at 0.6 m/s and must detect defects as small as 1.5 mm, measure width within ±2 mm and make a decision within 0.5 s. Human inspection alone may miss small defects because of fatigue and changing attention. Automated inspection can analyse every captured frame consistently and provide defect coordinates to the marker. An operator should supervise alarms and review uncertain cases.

1.4 Limitations and mitigation

- False positives from fabric texture: build a representative dataset, use texture-aware features and verify detections over neighbouring frames.
- Lighting changes from 500 to 3,000 lux: use an enclosed inspection hood, constant-current LED lighting, exposure control and periodic reference checks.
- Calibration drift or camera movement: use a rigid mount and calibration target; recheck calibration at planned intervals.
- New fabric types may change appearance: validate settings for each fabric type and maintain a labelled defect dataset.

TASK 2 – Levels of Computer Vision: Pipeline Design and Operation

2.1 Candidate pipeline diagrams

LOW-LEVEL PIPELINE

Camera → Preprocessing → Enhancement → Display / enhanced image

THREE-LEVEL PIPELINE

Camera → Preprocessing & enhancement → Feature extraction → Defect decision
(edges, texture, colour, shape)

DEEP-LEARNING PIPELINE

Camera → Resize/normalise → Trained neural network → Class + confidence + location

2.2 Inputs and outputs

Stage	Input	Operation	Output
Acquisition	Physical fabric and light	Capture image at calibrated scale	Raw image
Preprocessing	Raw image	Correct brightness, crop region, remove noise	Clean image
Enhancement	Clean image	Contrast adjustment, colour conversion, filtering	Defects made more visible
Feature extraction	Enhanced image	Measure edges, texture, colour, connected regions and shape	Numerical feature set
Decision	Features or image patches	Apply thresholds/rules or trained classifier	Defect type, confidence and coordinates

2.3 Required stages for defect types

- Holes: background/foreground contrast, thresholding, contour detection, area and shape checks.
- Broken threads: directional edge and line analysis, local texture comparison and detection of interrupted weave patterns.
- Oil stains: colour-space analysis and local contrast enhancement; controlled illumination is important because the stated contrast is only 5– 8 grey levels.
- Weaving irregularities: texture features, periodicity analysis and comparison with a reference region or known-good fabric.

2.4 Comparison

Approach	Complexity	Explainability	Hardware	Response
Low-level only	Low	High; each operation is visible	Low	Fast, but cannot make reliable defect decisions alone
Three-level	Medium	High to medium; features and rules can be inspected	Suitable for 4-core CPU	Can meet timing if image size and algorithms are controlled
Deep learning	High at training and inference	Lower; decisions may be harder to explain	Often benefits from GPU	May be fast on suitable hardware, but CPU inference and model size must be tested

2.5 Recommendation

Use a three-level pipeline on the available 4-core CPU. It provides acquisition, preprocessing, explicit feature extraction and a final decision while remaining explainable and relatively lightweight. Use OpenCV operations such as colour conversion, Gaussian filtering, adaptive thresholding, morphology, contours and texture measurements. Optimise the region of interest and process only necessary frames. A compact trained model may be evaluated later, but a GPU-dependent design should not be assumed.

TASK 3 – Image Representation Analysis

3.1 Representations

Representation	Information stored	Textile inspection suitability
RGB	Three channels: red, green and blue; each commonly 8 bits	Preserves original colour and is useful for colour variation and coloured stains, but brightness changes affect all channels.

HSV	Hue, saturation and value channels	Separates colour information (hue/saturation) from brightness (value). Useful for colour differences and some stains, but hue becomes unreliable for very low-saturation pixels.
Grayscale	One intensity channel	Lower memory and simpler processing. Useful for holes, thread edges and texture; may lose colour information needed for colour defects.

3.2 Defect-specific use

- Colour variations: RGB or HSV is preferred because grayscale discards hue information.
- Oil stains: use HSV and grayscale/local contrast together. The low 5– 8 grey-level contrast means controlled lighting and careful exposure are essential.
- Holes: grayscale thresholding and contour analysis can separate the hole from the fabric when the background contrast is sufficient.
- Broken threads: grayscale edge and texture analysis is efficient; colour channels can be retained for confirmation.

3.3 Memory and complexity

For an image of W × H pixels, an 8-bit grayscale image requires W× H bytes. An 8-bit three-channel RGB or HSV image requires approximately 3× W× H bytes. HSV conversion adds computation, but the three channels occupy the same approximate storage as RGB.

3.4 Selection

Use a combined representation: retain RGB at acquisition, convert to HSV for colour/stain analysis, and create a grayscale image for edge, texture and hole analysis. This preserves colour evidence while allowing efficient single-channel processing for structural defects.

TASK 4 – Connectivity Analysis

4.1 Definitions and neighbourhoods

For a centre pixel P=(x,y):

4-neighbours: (x-1,y), (x+1,y), (x,y-1), (x,y+1)

8-neighbours: the 4-neighbours plus (x-1,y-1), (x+1,y-1),
(x-1,y+1), (x+1,y+1)

4-connectivity: 8-connectivity:

. 1 . 1 1 1

1 P 1 1 P 1

. 1 . 1 1 1

m-connectivity: diagonal neighbours are connected only when their common 4-neighbours do not contain a foreground pixel.

4.2 Example binary image

1 0 0

0 1 0

0 0 1

The three foreground pixels touch only diagonally.

4-connectivity: 3 separate components.

8-connectivity: 1 component.

m-connectivity: 3 separate components in this example because each diagonal pair has a common 4-neighbour that is 0,

so the diagonal link is allowed; therefore all three pixels form 1 component.

Correction for the example: under the stated m-connectivity rule, the common 4-neighbours of each diagonal pair are background (0), so diagonal links are allowed and the three pixels form one component. If a common 4-neighbour were foreground, that diagonal connection would be disallowed.

4.3 Effect and recommendation

4-connectivity can split a diagonal scratch or irregular defect into several regions. 8-connectivity joins diagonal pixels and may merge nearby defects that only touch at corners. m-connectivity reduces ambiguous diagonal linking when common 4-neighbours are foreground. For small holes and irregular textile defects, use 8-connectivity for initial region grouping when continuity is important, then apply shape, distance and area checks to avoid merging separate defects. m-connectivity is a useful alternative when diagonal ambiguity causes false merging.

TASK 5 – Image Acquisition Analysis

5.1 Principles

Camera	Acquisition principle	Effect on moving fabric
Rolling shutter	Rows are exposed at slightly different times	Motion can skew or distort objects because the fabric moves while rows are captured.
Global shutter	All pixels expose at approximately the same instant	Freezes motion more effectively than rolling shutter; useful for area images of moving fabric.
Line scan	One line is captured at a time; motion of the fabric builds the 2D image	Naturally suited to continuous web inspection; requires encoder or accurate line-rate synchronisation.

5.2 Motion analysis

At 0.6 m/s, the fabric travels 0.6 mm in 1 ms. A rolling-shutter readout lasting several milliseconds can create visible displacement between the top and bottom of an image. A global shutter reduces this geometric distortion when a short exposure is used. A line-scan camera avoids frame gaps in continuous web inspection, but its line rate must be synchronised to conveyor movement.

5.3 Comparison and selection

Factor	Rolling shutter	Global shutter	Line scan
Moving-web image quality	Motion skew risk	Good with short exposure	Excellent when synchronised
Continuous coverage	Frame overlap/gaps must be managed	Frame overlap/gaps must be managed	Continuous strip acquisition
Installation	Simple	Moderate	Needs line-rate/encoder setup
Cost	Often low	Moderate to high	Moderate to high; optics and lighting matter

A line-scan camera is the preferred industrial choice for a continuous fabric web, especially across the 1,000 mm fabric width. Select sufficient cross-web pixels to resolve 1.5 mm defects and synchronise the line rate with an encoder. A global-shutter area camera is a practical lower-cost prototype option.

5.4 Timing and resolution note

The camera-to-marker travel time is distance/speed = 0.3 m / 0.6 m/s = 0.5 s. This is the total physical time available before the detected portion reaches the marker. The processing decision must be made in less than 0.5 s, preferably with a safety margin (for example, below 0.3 s), because communication and actuator response also take time.

TASK 6 – Sampling Analysis

6.1 Pixel counts and spatial sampling

Assumption: the image width spans the full 1,200 mm field of view. Spatial interval = 1,200 mm / number of horizontal pixels. Approximate pixels across a 1.5 mm defect = 1.5 mm / spatial interval.

Resolution	Total pixels	Sampling interval	Pixels across 1.5 mm
1280 × 1024	1,310,720 (1.31 MP)	1200/1280 = 0.9375 mm/px	1.5/0.9375 = 1.60 px
2448 × 2048	5,013,504 (5.01 MP)	1200/2448 = 0.4902 mm/px	1.5/0.4902 = 3.06 px
4096 × 3000	12,288,000 (12.29 MP)	1200/4096 = 0.2930 mm/px	1.5/0.2930 = 5.12 px

6.2 Adequacy and trade-off

A defect represented by only 1.6 pixels is difficult to detect reliably and may disappear depending on its orientation and sampling phase. About 3 pixels is a minimum practical representation for basic detection, while 5 pixels gives better shape evidence. Therefore, low resolution is inadequate; medium is borderline-to-acceptable for simple detection; high resolution provides the strongest detail.

Resolution	Approx. uncompressed RGB frame (3 bytes/pixel)	Processing/storage impact
1280× 1024	3.93 MB	Lowest; fast but insufficient defect sampling
2448× 2048	15.04 MB	Moderate; manageable with ROI and optimised CPU processing
4096× 3000	36.86 MB	High bandwidth, memory and processing demand

6.3 Selection

Select 2448 × 2048 as the initial balanced choice for the 4-core CPU, with controlled lighting and a calibrated lens. It gives about 3.06 pixels across a 1.5 mm defect. If validation shows that defect shape or the 2% missed-defect target cannot be achieved, move to high resolution or use a line-scan system with adequate cross-web sampling. The decision must be verified experimentally; pixel count alone does not guarantee detection.

TASK 7 – Quantization Analysis

7.1 Intensity levels

Quantization	Number of levels	Typical representation
8-bit	2^8 = 256 levels	0– 255
12-bit	2^12 = 4,096 levels	0– 4095
4-bit	2^4 = 16 levels	0– 15
6-bit	2^6 = 64 levels	0– 63

7.2 Low-contrast stain analysis

The oil-stain contrast is only 5– 8 grey levels. With 8-bit quantization, this difference can be represented directly if sensor noise, lighting variation and compression do not overwhelm it. A 12-bit sensor provides more intensity steps and can preserve subtle differences before conversion, but needs more data and processing. Reducing to 4– 6 bits risks mapping nearby values to the same quantized level, making a faint stain disappear.

7.3 Storage and processing

For the same pixel count, 12-bit data requires 1.5 times the raw bit storage of 8-bit data when tightly packed (often stored in 16-bit containers, which can double it). Four- or six-bit data reduces storage but sacrifices tonal detail. Processing 8-bit images is generally efficient on the available CPU.

7.4 Recommendation

Use an 8-bit processing pipeline as the baseline, with a camera capable of higher bit-depth capture if affordable and useful for the low-contrast stain. Preserve 10/12-bit raw data during acquisition where supported, then apply calibrated conversion and contrast enhancement. Do not reduce to 4– 6 bits before stain detection.

TASK 8 – Image Formation Analysis

8.1 Image formation

Light from the illumination source reflects from the fabric and passes through the lens to the camera sensor. The lens maps points on the fabric plane to image coordinates. At a 700 mm camera height, lens focal length, sensor size, lens distortion, camera tilt and fabric-plane position determine the scale and geometry of the image.

8.2 Perspective and width measurement

If the camera is tilted or the fabric is not parallel to the sensor, the apparent scale changes across the image. Lens distortion can also bend straight edges. These errors affect the measurement of the 1,000 mm fabric width and may exceed the required ±2 mm tolerance.

8.3 Calibration

- Calibrate intrinsic parameters (focal length and lens distortion) using a checkerboard or calibration grid.
- Use a planar calibration target placed at the fabric plane to estimate a homography and rectify perspective.
- Convert pixels to millimetres using known dimensions and verify with a traceable width reference.
- Fix the camera mechanically and repeat calibration after any movement or lens change.

8.4 Lighting variation

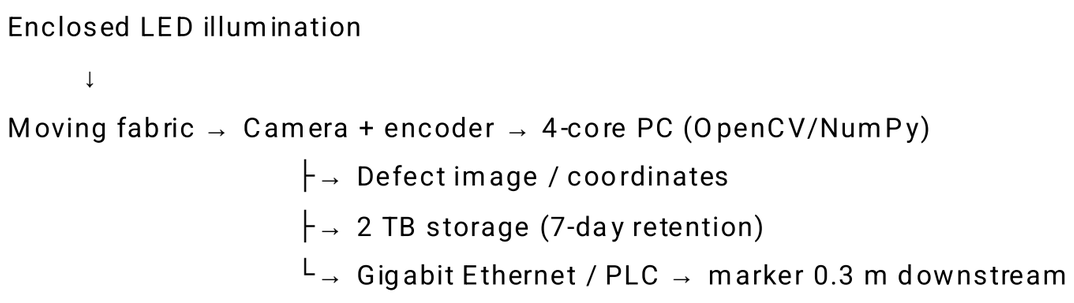
A change from 500 to 3,000 lux is a six-fold change in illumination. It can alter image brightness, contrast and threshold results. Oil stains with only 5– 8 grey levels of contrast are especially vulnerable. Use a light-controlled enclosure, diffuse or line lighting suited to the fabric texture, fixed exposure/gain and a reference white/grey check.

8.5 Recommendation

Use a rigidly mounted, calibrated camera with a low-distortion lens, camera axis perpendicular to the fabric plane, a planar geometric correction and enclosed constant-output LED illumination. Measure width after rectification and verify accuracy across the full 1,000 mm width. A telecentric lens may improve dimensional stability but must be assessed against cost and field-of-view requirements.

TASK 9 – Implementation and Performance Evaluation

9.1 Proposed system block diagram



9.2 Configuration comparison (planning estimates)

Configuration	Estimated cost	Resolution / hardware	Estimated processing	Assessment
Budget	₹ 55,000– ₹ 75,000	Entry-level area camera; basic LED; existing 4-core PC	0.2– 0.5 s depending on frame size	Lowest cost; lighting and small-defect performance may be marginal
Balanced	₹ 1,15,000– ₹ 1,45,000	Global-shutter or entry line-scan camera; 5 MP	Target 0.15– 0.30 s after ROI and code	Best balance; requires benchmark and

		class; controlled LED; encoder; existing PC	optimisation	acceptance testing
Premium	Above ₹ 1,50,000 (estimate)	High-resolution line-scan camera, advanced optics/lighting and upgraded processor	Potentially below 0.15 s with suitable hardware	Strong capability but exceeds budget and needs additional computing

Cost assumptions: estimates include camera, lens, lighting, mounting/enclosure, trigger/encoder interface, PLC or I/O, cabling and basic integration allowance. They exclude major conveyor modifications and taxes/vendor-specific pricing. Obtain quotations before procurement.

9.3 Defect detection and recording

- Holes: detect local intensity differences, threshold the image, find connected components and filter by area and shape.
- Broken threads: analyse directional edges and local texture; flag discontinuities compared with nearby weave patterns.
- Oil stains: use stable illumination, HSV/grayscale channels and local contrast; compare stain-like regions with known fabric background.
- Weaving irregularities: calculate local texture or periodicity features and compare with a reference pattern.
- For every defect, save timestamp, defect class, confidence, pixel coordinates, calibrated millimetre coordinates and a cropped evidence image.

9.4 Marker communication and timing

The defect's machine-direction coordinate is tracked using the conveyor encoder. At 0.6 m/s, the 0.3 m distance corresponds to 0.5 s. The PC sends the defect position and trigger timing to a PLC or industrial I/O module over a deterministic interface. The PLC compensates for actuator delay and fires the marker when the same fabric portion reaches it. Gigabit Ethernet can carry images and logs, but a time-critical trigger should use a reliable PLC/I/O signal rather than depend only on a general-purpose network message.

9.5 Storage estimate

For an illustrative 5 MP RGB frame of about 15 MB, storing every frame at 10 frames/s would require about 150 MB/s, or roughly 907 GB per hour. This is not practical for seven days on a 2 TB drive. Therefore, store defect crops, event images and metadata by default, and retain full frames only at a controlled lower rate or when a defect is detected. Apply a seven-day automatic deletion policy. Actual storage depends on frame rate, compression and defect frequency.

9.6 Evaluation and selection

Requirement	Budget	Balanced	Premium
0.5 s decision	May meet after optimisation; risk is high	Target is feasible; must be benchmarked	Likely feasible with suitable hardware
7-day retention	Only with event-based storage	Feasible with event-based storage	Feasible with storage policy
Missed defects ≤2%	Not assured; validate carefully	Target candidate; requires representative testing	Potentially strongest, but not guaranteed
Budget limit	Within budget	Within budget if procurement is controlled	Likely exceeds budget

Select the balanced configuration, keeping total procurement and integration at or below ₹ 1,50,000. It is the most practical combination of resolution, controlled illumination and CPU processing. The 2% missed-defect requirement cannot be guaranteed from specifications alone; conduct a trial with labelled samples across fabric types, speeds, lighting conditions and all defect classes. Measure recall = TP/(TP+FN) and require recall of at least 98% for the agreed defect set.

TASK 10 – Front-End Adaptation Challenge

10.1 Components that become inadequate

- Existing 1.5 mm sampling is insufficient for 0.4 mm defects unless spatial resolution is increased substantially.
- At 1.2 m/s, the fabric moves twice as fast; exposure must be shorter and line-scan rate/encoder frequency must double for the same spatial sampling along the fabric.
- Visible-only lighting/camera may not reveal stains that require UV illumination.
- The existing 4-core CPU may not process twice the production volume at the stricter 0.3 s decision deadline, especially at higher resolution.
- The existing marker timing and PLC logic must be recalculated for the faster line.

10.2 Sampling requirement

A 0.4 mm defect should ideally cover at least 4– 5 pixels for reliable shape evidence. Required sampling interval = 0.4 mm / 5 = 0.08 mm/pixel. Across 1,200 mm, required horizontal pixels = 1,200/0.08 = 15,000 pixels. Even a 4,096-pixel-wide camera gives only 1,200/4096 = 0.293 mm/pixel, or 0.4/0.293 = 1.37 pixels across the defect. Thus a much higher-resolution line-scan camera, narrower field of view/multiple cameras, or optical magnification is required.

At 1.2 m/s, to maintain 0.08 mm sampling in the machine direction, line rate = 1,200 mm/s / 0.08 mm = 15,000 lines/s. This is a demanding acquisition rate and produces a high data load.

10.3 UV imaging

Add a UV-A illumination source and a camera/sensor with suitable UV sensitivity and optical filtering. Some stains may fluoresce or show different reflectance under UV. The system should capture a visible channel and a UV channel, calibrate both channels and fuse their detections. UV safety shielding and appropriate exposure controls are required. The actual stain chemistry must be tested; UV does not make every stain visible.

10.4 Adaptation strategies

Strategy	Feasibility	Accuracy / speed	Cost and risk
Software-only changes	Low feasibility	Cannot recover detail absent from existing pixels or sensor spectrum; may optimise CPU time only	Low cost but high risk of missing 0.4 mm defects and UV stains
Upgrade selected components	Moderate	Replace camera/optics, add UV channel and lighting, encoder and optimised software; existing PC may remain a bottleneck	Cost controlled by selecting essential upgrades; integration risk moderate
Full redesign	Technically strongest	High-resolution multi-channel line-scan acquisition with high-speed processing can target new requirements	Highest cost and implementation risk; unlikely within unchanged budget and PC

10.5 Recommendation and updated block diagram

Visible LED + UV-A illumination

↓

High-resolution line-scan camera(s) + encoder

↓

Preprocessing / calibration / visible-UV fusion

↓

Optimised ROI and lightweight defect detection

↓

Defect coordinates + timestamp

↓

PLC / high-speed marker control

↓

Event storage + network reporting

The most practical strategy is a selective component upgrade, prioritising a high-resolution line-scan acquisition system, UV illumination/channel, encoder synchronisation and efficient ROI-based software. However, the unchanged PC and budget impose serious limits. A 15,000-pixel cross-web requirement and 15,000 lines/s machine-direction rate can create data throughput beyond what a 4-core CPU can reliably process within 0.3 s.

10.6 Requirements that cannot realistically be guaranteed

- 0.4 mm detection across the full 1,200 mm field of view cannot be guaranteed with the existing 4,096-pixel width; it needs substantially more sampling or multiple/narrower views.
- UV stain identification requires UV-capable optics/sensor and testing of the actual stain materials; software-only changes cannot create missing spectral information.
- Twice the production volume with a 0.3 s decision on the unchanged 4-core CPU is not a safe guarantee at the required resolution.
- The unchanged ₹ 1,50,000 budget may not cover the high-resolution line-scan camera, UV channel, optics, lighting and integration. The company must prioritise requirements or approve additional resources.

Therefore, the upgrade can improve capability, but all new targets cannot honestly be guaranteed without additional computing, acquisition bandwidth and budget. A pilot test should define achievable detection limits before accepting the export order.

Conclusion

A calibrated, controlled-illumination computer-vision system is suitable for textile inspection because it can detect and locate defects consistently during continuous production. For the original constraints, a three-level CPU-based pipeline with a balanced camera and event-based storage is recommended. A line-scan camera is best suited to continuous web inspection, while a 5 MP-class sampling choice provides a practical starting point for 1.5 mm defects. For the stricter export order, high-resolution line-scan acquisition and UV imaging are needed, and the unchanged PC and budget are unlikely to support every new requirement.

References and Tools

- Course assignment question paper: ITA05 – Computer Vision, Assignment 1 – CO1, SIMATS Engineering (provided source).
- OpenCV documentation: https://docs.opencv.org/
- NumPy documentation: https://numpy.org/doc/
- Gonzalez, R. C. and Woods, R. E., Digital Image Processing, Pearson – image sampling, quantization and image processing fundamentals.
- Szeliski, R., Computer Vision: Algorithms and Applications – image formation, calibration and computer vision pipeline concepts.

Software proposed: Python 3, OpenCV (cv2), NumPy. All cost, latency and detection-rate figures must be verified with actual hardware and fabric samples before industrial deployment.